MICROSOFT 365 COPILOT

Network Defense Intelligence Exercise Research and Validation Dossier

Evidence Before Architecture

Field	Value
Document ID	OAL-RESEARCH-NDIE-COPILOT-001
Version	v1.0 - Independent Research Execution
Date	November 4, 2026
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Source prompt	Microsoft Copilot Independent Research Prompt v1.1
Repository target	network-defense-intelligence-exercise
Status	Independent research report; not canon approval; not deployment authorization

Distribution notice
Prepared for cross-AI review and GitHub archival. This report separates sourced evidence, planning estimates, hypotheses, fictional canon, and recommendations. It contains no authorization for live testing or deployment.

Executive Decision Summary

The central defensive principle is worth a bounded research program, but the complete national-scale architecture described in the source dossier is not presently supported by evidence. Real-world practice supports decoys, adversary engagement, moving-target techniques, cyber ranges, digital twins, zero-trust architectures, and machine-speed vulnerability remediation as separate or partially integrated capabilities. Public evidence located in this review does not establish seamless migration of all hostile activity into a complete synthetic national infrastructure, universal end-to-end attribution, or a fixed three-second rotation that preserves state and availability at national scale.

The strongest defensible goal is narrower: determine whether selectively placed, authorized decoys and adaptive isolation can reduce unauthorized reachability and mission harm beyond matched baseline controls, while producing useful intelligence and tolerable operational overhead. CISA now explicitly describes cyber decoys as complements to Zero Trust for detecting post-compromise activity, including valid-credential and living-off-the-land behavior. MITRE Engage likewise frames deception as an intentional, iterative adversary-engagement process rather than a fire-and-forget product. [S1] [S2]

The most consequential weakness is not honeypot fingerprinting. It is coverage. An actor may already possess legitimate access, enter through a supplier or control plane, operate through a human or physical process, target an unmanaged or OT asset, or avoid the decoy altogether. Any claim that all attackers are captured or real infrastructure is fully secured is therefore not a measurable real-world end state unless the protected domain, observation window, and ground truth are tightly bounded.

Recommendation: GO to G1 research justification and a G2 isolated prototype plan; do not proceed to a live or sector pilot. The first empirical unit should be one consenting organizational enclave in a cyber range, not a national-scale shadow network. Compare ordinary controls, static decoys, adaptive decoys without rapid cycling, and adaptive decoys with randomized/event-driven cycling. Primary outcomes must be unauthorized real-asset reachability, mission impact, availability, and recovery, not attacker engagement alone.

Conditional decision
GO for a staged research program. REVISE the real-world interpretation. PAUSE any live pilot. STOP if isolated testing shows no measurable advantage over simpler controls, unacceptable benign-user harm, inadequate isolation, or unaffordable sustainment.

1. Source-Dossier Audit

The baseline dossier and the v1.1 Copilot prompt were retrieved as user-owned Microsoft 365 files. The dossier describes the fixed fictional sequence as detection, mapping, neutralization of access to real assets, synthetic generation, silent migration, approximately three-second cycling, black-line collapse, and transfer. It also states that the dossier is pending ONESTAMP and separates canon facts from draft suggestions. The v1.1 prompt requires an adversarial, evidence-led execution rather than confirmation research.

Material source issues

- The approval notice says the dossier is pending ONESTAMP, while later language calls some elements fixed or non-negotiable. For this report, they are treated as fictional design constraints, not empirical facts.
- The phrase “every active attacker” has an unknowable denominator in a real environment. A measurable substitute is the proportion of ground-truth test campaigns detected, diverted, or prevented inside a bounded observation window.

- The sequence contains a transition seam: if genuine reachability is already neutralized, deception adds observation and delay rather than first-order protection; if it is not yet neutralized, migration creates an exposure interval.
- The black-line disappearance is a visualization event, not proof of isolation, attribution, campaign termination, or identification of a person.
- The transfer package is simultaneously a defensive product, intelligence repository, update pathway, and potential supply-chain target. It should be treated as a high-value protected system.
- The five-tier framework requires an authoritative taxonomy before engineering use because the dossier uses Tier 0 and Tier 1 inconsistently for critical assets.
- The dossier's role model gives the Architect sole full visibility and authority. That creates a continuity and single-point-of-failure problem that requires separation of duties and independent safety authority.

2. Formal Research Goal

Primary objective: test whether authorized selective deception plus adaptive isolation measurably reduces unauthorized reachability and mission impact compared with matched conventional controls against adversaries who may engage, recognize, avoid, escape, or bypass the deceptive surface.

Secondary objectives

- Measure detection and isolation time.
- Measure benign-user diversion, latency, availability, and analyst workload.
- Evaluate intelligence value from adversary engagement.
- Test whether rotation policy adds value beyond static or event-driven decoys.
- Identify coverage gaps and independent controls needed when deception does not engage.
- Evaluate safe transfer, rollback, and receiving-operator operation.

Falsification criteria

- No statistically or operationally meaningful reduction in real-asset reachability versus simpler controls.
- Availability, latency, or analyst workload costs outweigh the measured benefit.
- Adversaries regularly avoid the deceptive surface without additional warning.
- Containment cannot be independently verified.
- Attacker-derived data contaminates trusted policy or update artifacts.
- The architecture requires authority or telemetry unavailable to the intended operator.

3. Why the Research Matters

NIST defines cyber resiliency as the ability to anticipate, withstand, recover from, and adapt to adverse cyber conditions, and it treats resiliency as a systems-engineering and lifecycle problem rather than a single product. [S3]

CISA's September 2026 decoy guidance states that organizations struggle to detect actors using legitimate credentials, native tools, and living-off-the-land techniques; it positions decoys as complementary to Zero Trust, with high-fidelity alerts and post-compromise visibility. [S1]

MITRE Engage provides a framework for planning and discussing adversary engagement; MITRE emphasizes that deception is a process and recommends avoiding hack-back. [S2] [S4]

The research is justified because these sources support a real defensive problem: prevention can fail, post-compromise activity can resemble authorized behavior, and defenders need additional ways to detect, misdirect, and learn. The research is not justified as a direct leap to national-scale synthetic replication. The incremental-value question must remain open.

4. Real-World Technology Baseline

Demonstrated or operationally recognized

- Decoys and concealment/misdirection are recognized defensive controls. NIST SC-26 describes decoys designed to attract, detect, deflect, and analyze attacks; SC-30 describes concealment, randomness, relocation, and misleading information, while noting added complexity and management overhead. [S5] [S6]
- Zero Trust focuses on protecting resources rather than trusting network location. That makes it an essential baseline and an independent control when deception is bypassed. [S7]
- MITRE D3FEND provides a knowledge graph of defensive countermeasures, and MITRE Engage provides adversary-engagement planning tools. [S8] [S2]
- Cyber ranges provide safe, simulated environments for realistic exercises, assessment, and testing. [S9]
- DARPA's 2016 Cyber Grand Challenge demonstrated automated vulnerability discovery and patching in a purpose-built air-gapped competition, not national network deception. [S10]
- DARPA's 2025 AI Cyber Challenge demonstrated AI systems finding and patching vulnerabilities in critical-infrastructure-relevant open-source software. [S11]
- Digital-twin security research exists at bounded sector scale; DOE-funded DISCOVER is a three-year, $4.8 million project to test software and firmware through an OT digital twin. [S12]

Partially demonstrated or research-stage

- Moving-target defenses can manipulate addresses, configurations, diversity, or timing, but surveys identify fragmented research and performance/security tradeoffs. [S13] [S14]
- Cyber deception surveys document honeypots, honeytokens, traffic redirection, dynamization, and integrated deception but also emphasize detection and bypass by professional adversaries. [S15] [S16]
- Game-theoretic work can estimate value of deception and break-even conditions relative to matched baselines, but modeling results are not the same as operational national-scale validation. [S17]

Unsupported at proposed scale

- Complete national synthetic infrastructure generated on demand with production-equivalent fidelity.
- Transparent migration of arbitrary established authenticated sessions across heterogeneous organizations.
- Reliable intent-level inference with negligible false positives at national scale.
- Universal capture or attribution of every active attacker.
- A fixed three-second rotation across routing, identity, keys, services, and telemetry without unacceptable availability or state-consistency effects.
- Safe transfer of a self-adapting national deception engine through an ordinary reboot/update cycle.

5. Prior Exercises and Comparable Programs

Cyber Storm and Locked Shields establish that large-scale cyber exercises are real, valuable, and multidisciplinary. They do not demonstrate the proposed absorption architecture.

CISA's Cyber Storm VIII involved more than 2,000 participants and tested response to multiple concurrent incidents, emphasizing coordination and information sharing. Cyber Storm IX involved more than 2,200 participants and focused on cloud shared responsibility and multi-sector response. [S18] [S19]

Locked Shields is described by NATO CCDCOE as a large international live-fire defensive exercise that combines technical, operational, strategic, legal, and communications considerations. [S20]

DOE CESER frames exercises as objectives-based mechanisms to validate capabilities, measure resilience, identify gaps, and create improvement actions. That process is directly transferable to NDIE. [S21]

Transferable lesson: exercise value comes from explicit objectives, injects, evaluators, after-action reporting, coordination, and improvement planning. Non-transferable inference: success in a cyber range does not prove universal real-world visibility, seamless cross-owner routing control, or national-scale containment.

6. Capability and Dependency Assessment

The architecture should be understood as a chain. Failure of an early dependency invalidates later claims. Selective decoys can operate with local administrative control; transparent migration and national coverage require much broader authority, state, and trust.

- Observe and classify: depends on telemetry quality, sensor coverage, identity context, and model calibration.
- Map reachable paths: depends on current asset inventory, routing, identity, cloud control planes, and third-party connections.
- Prevent access to real assets: depends on segmentation, identity enforcement, policy engines, and fail-safe behavior. This is a conventional-control function, not a deception-only function.
- Instantiate decoys: feasible locally through VMs, containers, accounts, files, services, and simulated data; fidelity and maintenance cost rise with scope.
- Redirect eligible activity: practical at controlled gateways or application layers; arbitrary established encrypted/authenticated sessions are much harder.
- Monitor and isolate: requires trustworthy logs, egress controls, independent verification, and incident-response integration.
- Adapt: requires change control, poisoning resistance, validation, human oversight, and rollback.
- Transfer: requires provenance, signing, inspection, operator acceptance, key management, rollback, and revocation.

7. Adversarial Analysis: Recognition, Avoidance, Escape, and Bypass

Recognition

An adversary may infer deception from impossible topology, stale organizational details, synthetic user behavior, clock or latency anomalies, inconsistent certificates, missing external dependencies, or regular cycling. The defense must measure recognition rate rather than assume invisibility.

Avoidance

A patient actor may avoid attractive assets, use only legitimate workflows, wait for maintenance windows, or use sacrificial probes. Deception still has value if avoidance itself creates a reliable signal, but that must be demonstrated.

Escape

A compromised decoy can become a staging point if egress, orchestration, identity, or management planes connect back to production. Independent isolation verification is required; apparent engagement is not enough.

Coverage bypass

The most serious classes include pre-positioned access, trusted identities, supply-chain compromise, cloud or vendor control planes, unmanaged systems, physical maintenance paths, insiders, OT processes, and attacks on the defender's sensors or update mechanism. These classes make "fully secured" untenable outside a bounded test domain.

8. Three-Second Hypothesis Assessment

No source reviewed establishes three seconds as a generally optimal reconfiguration interval. MTD performance research explicitly treats reconfiguration rate as a tradeoff constrained by performance and stability. [S14]

Three seconds can remain a narrative metronome, but the empirical program should treat it as one experimental condition. The safest interpretation is to decouple the visible three-second pulse from wholesale changes to every network, cryptographic, identity, and application state.

- Compare static, fixed 3-second, slower fixed, randomized, and event-driven policies.
- Specify exactly what changes. Address rotation, route translation, service personality, key material, and session context must be evaluated separately.
- Measure dropped sessions, latency, jitter, cache inconsistency, logging gaps, forensic continuity, compute cost, recognition rate, and real-asset reachability.
- Reject the 3-second engineering interpretation if it increases detection, availability loss, or defender uncertainty without reducing compromise.

9. Longitudinal Evolution

The likely strategic trajectory is not permanent invisibility. Early deployments may gain information advantage against actors unfamiliar with the environment. Repeated exposure increases the chance of recognition, avoidance, planted misinformation, and attacks on the transfer/update pathway. Defenders then face maintenance, doctrine leakage, configuration drift, staff turnover, and escalating fidelity costs.

1. Generation 0: unknown defense; measure baseline engagement and protection.
2. Generation 1: known use of decoys; measure avoidance and displacement.
3. Generation 2: broad characteristics known; measure fingerprint testing.
4. Generation 3: adversary runs reality checks; measure external-consistency failures.
5. Generation 4: adversary poisons captured behavior; measure model and policy contamination.
6. Generation 5: reciprocal deception; prioritize independently verifiable mission state and trusted ground truth.

The system should be reevaluated whenever recognition rate rises, maintenance exceeds planned thresholds, benign-user harm increases, operator trust declines, or the defense no longer outperforms simpler controls.

10. Defender Learning and Poisoning Risk

Captured behavior should be treated as untrusted evidence. It may include deliberate false artifacts, sacrificed infrastructure, benign intermediaries, or behavior designed to shape future classification. Automatic conversion of captured behavior into production policy is not justified.

- Use quarantine datasets and provenance.
- Separate collection from policy generation.
- Require independent validation against clean and adversarial holdouts.
- Stage changes in the range before production.
- Require human approval for high-impact rules.
- Maintain rollback, model/version lineage, and contamination indicators.

11. Information Flow, Authority, and Continuity

The source architecture's single full-visibility Architect model is narratively strong but operationally fragile. A real program should separate mission authority, safety authority, operations, forensic analysis, model governance, and release approval.

- No single AI should autonomously divert production traffic, update trusted policy, and certify its own success.
- An independent observer should maintain ground truth during experiments.
- A safety authority should be able to halt the test without relying on the deception engine.
- Forensic teams should receive the minimum production context necessary.
- Receiving operators must be able to inspect, reject, roll back, and safely degrade the transfer artifact.
- Continuity plans must address loss, compromise, coercion, or unavailability of the Architect/operator.

12. Experimental Program

Phase 0 - Evidence and design

Complete the source audit, threat model, asset boundary, metrics, legal review, and range design. No live testing.

Phase I - Isolated prototype

One synthetic enterprise enclave; known ground truth; no external connectivity. Test decoys, isolation, rotation policies, logging, and safe termination.

Phase II - Controlled cyber range

Human red/blue teams with matched scenarios. Compare baseline controls, static deception, adaptive deception without rapid cycling, and adaptive cycling policies.

Phase III - Repeated-adversary study

Separate participant cohorts and unseen scenarios. Measure learning, recognition, avoidance, and poisoning attempts.

Phase IV - Multi-organization federation

Simulate fragmented ownership, shared identity, cloud services, and partner boundaries. Do not use production traffic.

Phase V - Limited opt-in pilot

Consider only after replicated range advantage, signed authority, independent safety review, rollback success, and operator acceptance. Production deployment is not an automatic endpoint.

13. Measurement Framework

Primary measures

- Unauthorized reachability of designated real assets.
- Verified compromise rate under ground truth.
- Mission availability and safety impact.
- Recovery time and completeness.

Secondary measures

- Time to detection and verified isolation.
- Engagement coverage and non-engagement rate.
- False diversion of benign users.
- Deception recognition rate.
- Latency, session failure, and service availability.
- Analyst workload and alert quality.
- Forensic value scored against predefined questions.
- Compute, network, storage, and maintenance cost.

Every rate requires a denominator and observation window. Zero observed compromise must be reported as zero observed under the test conditions, not proof of zero risk.

14. Simulation-to-Reality Limits

- Simulation can validate logic, instrumentation, metrics, failure handling, and controlled comparisons.
- Emulation can validate interactions with real software and selected hardware.
- Human studies can test recognition and decision behavior under controlled rules.
- An opt-in pilot can test organizational integration and operational burden within one authorized boundary.
- No ethical test can establish universal national visibility or guarantee capture of unknown actors.
- A successful range cannot prove cross-jurisdiction authority, real-world attribution, or absence of unmodeled bypasses.

15. Safety Case

- Owned, isolated infrastructure only until explicit later authorization.
- No routing to third-party or production systems.
- Egress controls enforced outside the system under test.
- Synthetic data clearly marked and recoverable by authorized evaluators.
- Independent kill switch and safety authority.
- Stop for isolation loss, unexpected external traffic, privacy breach, unavailable rollback, or unacceptable mission disruption.
- Evidence preservation and post-test sanitization plan.
- No hack-back or modification of attacker-controlled systems. MITRE Engage explicitly advises engagement without hack-back; DOJ describes CFAA analysis as complex and authorization-dependent. [S4] [S22]

16. Transfer-Artifact Assurance

The transfer artifact should not be delivered as an opaque automatic update. Candidate models should be compared: inspectable package, limited appliance, federated service, signed playbook/signature bundle, escrowed runtime, or no transfer of the core.

- Software bill of materials and provenance.
- Reproducible or independently verifiable build.
- Cryptographic signing and key governance.
- Receiving-operator acceptance tests.
- Configuration validation and least privilege.
- Staged rollout, rollback, revocation, and safe mode.
- No automatic ingestion of untrusted attacker data into production policy.
- Continuity and exit plan if the original provider becomes unavailable.

17. Cost Model

No public source located supports a defensible national-scale price. The ranges below are Copilot planning estimates, not vendor quotes or awarded budgets. They are calculated from explicit staffing and infrastructure assumptions and should be replaced with quotations during procurement planning.

Comparable evidence: DOE's DISCOVER digital-twin project carries $4.8 million total funding over three years; AIxCC offered $29.5 million in aggregate prizes across a challenge ecosystem. These are scale markers, not direct cost analogues. [S12] [S23]

Principal cost drivers are labor, environment fidelity, OT/hardware-in-the-loop, telemetry retention, independent assessment, secure facilities, partner integration, and sustainment. Rotation frequency may raise compute and state-management cost but should be measured, not assumed.

18. Funding Pathways and Applicant Readiness

The strongest near-term pathway is a university-led or qualified small-business partnership for a bounded research study. A personal concept alone is unlikely to satisfy institutional, technical, compliance, and contracting expectations.

- NSF SaTC 2.0 is an active program supporting interdisciplinary security, privacy, resilience, and trust research. Official pages identify annual target dates and current solicitation NSF 25-515. [S24]
- DHS S&T lists a Long Range Broad Agency Announcement with a May 31, 2029 due date and identifies eligible groups including businesses, academia, national labs, and recognized R&D organizations. [S25]
- DHS SBIR is for qualified U.S. small businesses with fewer than 500 employees; its official page states typical Phase I funding up to $175,000 and Phase II funding of $1.0M-$1.5M. [S26]
- DARPA uses BAAs as a primary funding vehicle and states that BAAs may provide contracts, grants, or cooperative agreements without guaranteeing an award. Current opportunities must be checked on official DARPA and SAM.gov pages. [S27]
- SBIR.gov describes America's Seed Fund as nondilutive funding for U.S. small businesses, with phase ranges shown on its site; agency-specific topics and current authorization must still be verified before application. [S28]
- DOE CESER and national-laboratory partnerships are plausible for energy-sector digital-twin or OT testbeds, as illustrated by the DISCOVER award, but no commitment or open matching call was verified for this exact project. [S12]

Recommended first-dollar sequence

7. Complete the evidence dossier and a two-page neutral concept note.
8. Secure an academic cybersecurity or systems-engineering principal investigator, or form/qualify a small business if commercialization is intended.
9. Define one bounded use case and a safe cyber-range protocol.
10. Pursue a small feasibility or seed award before requesting a large pilot.
11. Require G1 evidence before G2 funding and replicated G2 results before any G3 request.

19. Legal, Ethical, and Governance Findings

This report is not legal advice. Deployment would require qualified counsel and written authorization. The DOJ CFAA manual describes cyber cases as legally and technically complex and emphasizes coordination and authorization issues. DOJ's 2022 good-faith research policy protects qualifying research from charging decisions but does not create general permission to test systems without authorization. [S22] [S29]

- Operate only within owned or expressly authorized systems.
- Separate internal deception from actions affecting third-party systems.
- Minimize and bound collection of innocent or partner traffic.
- Define retention, access, disclosure, and deletion rules.
- Use confidence-weighted attribution and prohibit automatic accusation.
- Provide independent oversight, audit logs, appeal/correction channels for misclassification, and responsibility for automated errors.
- Assess cross-border data, sector regulation, procurement, export, intellectual-property, and human-subject issues before each phase.

20. Comparative Alternatives and Opportunity Cost

Deception should complement, not replace, foundational controls. NIST Zero Trust removes implicit trust based on network location and authenticates/authorizes before resource access; CISA explicitly frames decoys as complementary to Zero Trust. [S7] [S1]

The G2 experiment should allocate comparable engineering effort to four treatments. If segmentation, identity controls, and high-quality monitoring achieve nearly the same protection with lower burden, the more complex adaptive system should not advance.

21. Independent Case Against Funding

- The architecture combines many individually difficult capabilities and may fail at the seams.
- It assumes visibility and authority that do not exist across diverse national infrastructure owners.
- National fidelity and three-second mutation may create unsustainable cost and availability risk.
- Adversaries may bypass the system through trusted identities, suppliers, control planes, or physical processes.
- The transfer package creates a high-value supply-chain and governance target.
- Simpler controls may provide most of the benefit.
- Successful engagement can be mistaken for successful protection.

This case would weaken if a controlled comparative study showed a replicated, material reduction in real-asset compromise with manageable overhead and no unacceptable bystander or governance harm.

22. Independent Case for a Limited Study

- Official guidance recognizes decoys and adversary engagement as useful components of defense-in-depth. [S1] [S2]
- Current literature identifies open questions about integration, quantified effect, operational cost, and adaptive deception. [S16]
- Cyber ranges provide a safe environment for testing new ideas. [S9]
- A bounded study can determine whether the distinctive combination of adaptive deception and verified isolation adds value over matched alternatives without exposing production assets.

The smallest defensible study is a single-enclave cyber-range experiment with four matched conditions and ground-truth attack campaigns, including non-engagement and bypass cases.

23. Required Changes Before Moving Forward

12. Replace universal capture with bounded, measured outcomes.
13. Replace purpose-reading with observable indicators and calibrated uncertainty.
14. Replace complete national replication with selective, pre-authorized deception zones for empirical work.
15. Treat three seconds as a tested policy variable and narrative pulse, not a universal engineering constant.
16. Replace black-line disappearance with audited state transitions: suspected, observed, diverted, isolation verified, visibility lost, resolved, unknown.
17. Require independent safety authority and separation of duties.
18. Make coverage-bypass experiments mandatory.
19. Quarantine attacker-derived data from direct policy updates.
20. Replace automatic reboot handoff with signed, inspected, staged, reversible transfer.
21. Keep canon, evidence, hypotheses, estimates, and decisions in separate repository tracks.

24. Decision Gates and Next Actions

Now - G0/G1

- Resolve protected mission, bounded asset class, institutional owner, and acceptable disruption.
- Create the claim ledger, source register, and authoritative asset taxonomy.
- Select a cyber-range partner or research institution.
- Draft rules of engagement, safety case, and data-governance plan.
- Convert this report into GitHub source Markdown and CSV registers.

Next - G2

- Build the isolated prototype.
- Run matched baseline/static/adaptive/cycling experiments.
- Publish negative and inconclusive results.
- Commission independent technical and legal review.

Later - conditional G3/G4

- Consider multi-organization federation only after replicated advantage.
- Consider an opt-in operational pilot only after successful rollback, safety, and operator-acceptance evidence.
- Do not plan a national deployment until authority, ownership, cost, and sustained reliability are independently demonstrated.

25. Readiness Assessment

Dossier v1.1 readiness
Ready for a research-oriented dossier revision and canon decision sprint. Not ready for a technical specification that assumes national-scale operability. Not ready for live testing or partner-agency transfer.

Confidence: moderate for the conclusion that bounded cyber-deception research is justified; high that the complete national-scale claim is unsupported by public evidence reviewed; low for any national-scale cost estimate because scope, authority, and architecture remain undefined.

26. Canon Translation

Empirical conclusion
Selective deception is plausible and recognized; universal national absorption is not established.

Engineering implication
Build a federated, bounded system-of-systems concept with explicit blind spots and independent controls rather than one omniscient layer.

Canon implication
The locked sequence can remain, but its real mechanism should include pre-positioned coverage, selective rendering, uncertain attribution, audited capture states, and a cost or limitation.

Narrative opportunity
The strongest failure is an orthogonal success by the adversary while the visible network-layer attack is convincingly trapped. This preserves the Architect's competence while revealing the limits of a system that only controls channels it can observe.

Author decision required
Whether the three-second pulse is literal actuation, interface rhythm, or both; whether "fully secured" is character perception or objective narrator fact; what the transfer artifact contains; and which canonical failure cost is accepted.

27. Remaining Unknowns

- Exact protected mission and first asset class.
- Institutional owner and authority.
- Definition of capture in canon and empirical work.
- Whether any AI component may autonomously divert traffic.
- Acceptable benign-user impact.
- Data-retention boundary.
- Public/private GitHub split.
- Budget ceiling and applicant entity.
- Partner-agency identity and inspection rights.
- Authoritative OAL asset-tier taxonomy.

Visual Appendix A - Capability Maturity Matrix

Capability	Assessment	Key limitation
Decoys/honeytokens	Demonstrated	Local scope; maintenance and isolation required
Adversary engagement planning	Demonstrated framework	Requires goals, policy, adaptation, and oversight
Moving-target reconfiguration	Research/partial deployment	Performance, stability, and management tradeoffs
Cyber-range national-crisis exercise	Demonstrated exercise class	Tests response, not universal containment
OT/critical-infrastructure digital twin	Bounded research deployments	Model fidelity and external validity
Automated vulnerability remediation	Demonstrated in competitions	Different problem from redirecting live campaigns
Selective session redirection	Feasible under controlled architectures	Depends on gateway/application control
Arbitrary established-session migration	Insufficient evidence	Encryption, identity, application state, cross-owner control
National synthetic infrastructure	Unsupported	Scale, authority, cost, fidelity, partial observability
Universal attacker capture/attribution	Unsupported	Unknown denominator and indirect infrastructure

Visual Appendix B - Bypass and Control Matrix

Bypass class	Failed assumption	Independent control	Residual risk
Pre-positioned access	Attack begins at monitored ingress	Asset inventory, hunting, identity controls	Dormant activity may remain invisible
Valid identity/insider	Hostility maps to anomalous endpoint	Least privilege, PAM, behavioral review	Malicious action may resemble work
Supply chain/update	Trusted software is safe	Signing, provenance, SBOM, staged update	Trusted channel can still be compromised
Cloud/vendor control plane	Defender owns relevant path	Contracted logging, federation, independent policy	Fragmented authority
OT/physical maintenance	Traffic crosses routed overlay	Physical access control, segmented OT, safety systems	Cyber layer cannot see all actions

Sensor/orchestrator compromise	Telemetry and policy are trustworthy	Independent monitors, attestation, separation of duties	Compromised root invalidates conclusions
Non-engagement	Adversary seeks bait	Coverage analytics and hardened real assets	No deception telemetry
Resource exhaustion	Procedural rendering stays bounded	Quotas, graceful degradation, priority policy	Adversary may force fidelity collapse
Transfer artifact compromise	Update channel is trusted	Signing, inspection, rollback, acceptance tests	High-value concentration

Visual Appendix C - Experimental Comparison Matrix

Arm	Description	Purpose
A	Conventional controls only	Establish protection baseline
B	Static decoys plus same conventional controls	Measure value of basic deception
C	Adaptive decoys without rapid cycling	Measure adaptation value
D	Adaptive decoys with randomized/event-driven cycling	Measure policy value
E	Adaptive decoys with fixed three-second condition	Test canon interval against alternatives

Visual Appendix D - Illustrative Cost Model

Stage	Illustrative scope	Low	Base	High	Confidence
G0-G1 feasibility study	6-9 months; 4-7 FTE-equivalent; literature, architecture, legal, threat model	$0.45M	$0.85M	$1.50M	Moderate
G2 isolated prototype	9-15 months; 8-15 FTE; isolated range, decoys, telemetry, independent test	$1.8M	$3.8M	$7.0M	Low-Moderate
G2-G3 comparative range study	12-24 months; 15-30 FTE; repeated red/blue	$5M	$10M	$20M	Low

	cohorts, OT option				
G3 multi-organization research demo	18-30 months; 30-60 FTE; federated synthetic environments	$15M	$30M	$60M	Low
G4 national operation	Undefined authority, scope, infrastructure, sustainment	Not responsibly estimable	Not responsibly estimable	Not responsibly estimable	Insufficient evidence

Planning assumptions: loaded annual labor rates of approximately $180,000-$300,000 per FTE-equivalent depending on specialization and secure-facility overhead; infrastructure and independent evaluation scaled by phase; 20%-35% contingency in higher-risk phases. These are Copilot assumptions, not sourced salary quotations.

Visual Appendix E - Funding Pathway Matrix

Pathway	Fit	Applicant readiness	Current evidence
NSF SaTC 2.0	High for foundational/interdisciplinary research	Typically institution-led proposal capacity required	Official active solicitation page located
DHS S&T LRBAA	Potential for operational homeland-security need	Qualified organizational proposer and strong mission fit	Official page lists May 31, 2029 due date
DHS SBIR	High for productizable bounded prototype	U.S. small business under 500 employees	Official phase guidance located
DARPA BAA	Potential for high-risk breakthrough research	Strong technical team; opportunity-specific compliance	Official BAA mechanism and current opportunities pages located
DOE CESER / labs	High for energy/OT digital twin version	University/lab/industry consortium likely	Comparable DISCOVER award located
Industry critical-infrastructure consortium	Potential pilot/in-kind support	Needs consenting operator and governance	No specific commitment verified
Venture capital	Poor fit for open-ended evidence program; better for product	Company, market, IP, and commercialization plan	No project-specific investor verified

Visual Appendix F - Decision Gate Matrix

Gate	Required evidence	Decision
G0 Concept clarity	Mission, owner, boundary, threat model, definitions, metrics	GO/REVISE before engineering

G1 Research justification	Comparable evidence, incremental-value hypothesis, bypass model	Fund study / narrow / stop
G2 Safe prototype	Isolated range, safety case, baseline arms, evaluator	Run controlled prototype only
G3 Controlled pilot	Replicated advantage, manageable harm, rollback, authority	Seek opt-in partner
G4 Scale	Independent review, sustained reliability, cost-benefit, legal authority	Scale only inside approved boundaries

Visual Appendix G - Evidence Register

ID	Organization	Title	Date	Evidence type
S1	CISA	Using Cyber Decoys to Strengthen Detection and Response	2026-09-16	Primary guidance
S2	MITRE	MITRE Engage	current	Primary framework
S3	NIST	SP 800-160 Vol. 2 Rev. 1 Developing Cyber-Resilient Systems	2021-12	Primary standard/guidance
S4	MITRE	Engage framework impact story	2022-02-28	Primary organization description
S5	NIST control reference	SC-26 Decoys/Honeypots	current control reference	Secondary rendering of NIST control
S6	NIST control reference	SC-30 Concealment and Misdirection	Rev. 5.2.0 reference	Secondary rendering of NIST control
S7	NIST	SP 800-207 Zero Trust Architecture	2020-08	Primary guidance
S8	MITRE	D3FEND Matrix and Resources	current	Primary framework
S9	NIST NICE	The Cyber Range: A Guide	2023-09	Primary guidance
S10	DARPA	Cyber Grand Challenge	2016	Primary program record
S11	DARPA	AI Cyber Challenge results	2025-08-08	Primary program result
S12	NYU Tandon / DOE award report	DISCOVER digital twin project	$4.8M; three years; posted 2024-03-20	University primary announcement
S13	Sun et al.	Survey on Moving Target Defense	2023	Peer-reviewed survey

S14	Connell, Menasce, Albanese	Performance Modeling of Moving Target Defenses	2017	Conference paper
S15	Javadpour et al.	Comprehensive survey on cyber deception techniques	2024	Peer-reviewed survey
S16	Zhang and Thing	Three Decades of Deception Techniques	2021	Preprint review
S17	Shahin, Maghanaki, Chen	Evaluating Synthetic Cyber Deception Strategies Under Uncertainty	2026-03-10	Peer-reviewed modeling study
S18	CISA	Cyber Storm VIII After-Action Report	2022-08	Primary after-action report
S19	CISA	Cyber Storm IX National Cyber Exercise	2024-04	Primary exercise record
S20	NATO CCDCOE	Locked Shields	current	Primary exercise record
S21	DOE CESER	Exercises and Training	current	Primary program guidance
S22	U.S. DOJ	Justice Manual 9-48.000 CFAA	current	Primary legal-policy source
S23	DARPA	AIxCC program record	$29.5M cumulative prizes	Primary program record
S24	NSF	SaTC 2.0 / NSF 25-515	active page accessed 2026-11-04	Primary funding source
S25	DHS S&T	Funding and Partnership Opportunities	updated 2026-08-31	Primary funding source
S26	DHS S&T	Small Business Innovation Research Program	current	Primary funding source
S27	DARPA	Office-wide Broad Agency Announcements	current	Primary funding source
S28	SBA	SBIR.gov America’s Seed Fund	current	Primary funding source
S29	U.S. DOJ	2022 CFAA charging policy announcement	2022-05-19	Primary policy announcement

Appendix H - Source URLs

S1 - CISA: Using Cyber Decoys to Strengthen Detection and Response (2026-09-16)

https://www.cisa.gov/resources-tools/resources/using-cyber-decoys-strengthen-detection-and-response

S2 - MITRE: MITRE Engage (current)
https://engage.mitre.org/

S3 - NIST: SP 800-160 Vol. 2 Rev. 1 Developing Cyber-Resilient Systems (2021-12)
https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final

S4 - MITRE: Engage framework impact story (2022-02-28)
https://www.mitre.org/news-insights/impact-story/mitre-engage-framework-and-community-cyber-deception

S5 - NIST control reference: SC-26 Decoys/Honeypots (current control reference)
https://csf.tools/reference/nist-sp-800-53/r4/sc/sc-26/

S6 - NIST control reference: SC-30 Concealment and Misdirection (Rev. 5.2.0 reference)
https://csf.tools/reference/nist-sp-800-53/r5/sc/sc-30/

S7 - NIST: SP 800-207 Zero Trust Architecture (2020-08)
https://csrc.nist.gov/pubs/sp/800/207/final

S8 - MITRE: D3FEND Matrix and Resources (current)
https://d3fend.mitre.org/

S9 - NIST NICE: The Cyber Range: A Guide (2023-09)
https://www.nist.gov/document/cyber-range

S10 - DARPA: Cyber Grand Challenge (2016)
https://www.darpa.mil/research/programs/cyber-grand-challenge

S11 - DARPA: AI Cyber Challenge results (2025-08-08)
https://www.darpa.mil/news/2025/aixcc-results

S12 - NYU Tandon / DOE award report: DISCOVER digital twin project ($4.8M; three years; posted 2024-03-20)
https://engineering.nyu.edu/news/nyu-tandon-tapped-us-department-energy-fortify-critical-infrastructure

S13 - Sun et al.: Survey on Moving Target Defense (2023)
https://doi.org/10.3390/app13095367

S14 - Connell, Menasce, Albanese: Performance Modeling of Moving Target Defenses (2017)
https://mason.gmu.edu/~wconnel2/PerformanceModeling1.pdf

S15 - Javadpour et al.: Comprehensive survey on cyber deception techniques (2024)
https://doi.org/10.1016/j.cose.2024.103792

S16 - Zhang and Thing: Three Decades of Deception Techniques (2021)
https://arxiv.org/abs/2104.03594

S17 - Shahin, Maghanaki, Chen: Evaluating Synthetic Cyber Deception Strategies Under Uncertainty (2026-03-10)
https://doi.org/10.3390/s26061748

S18 - CISA: Cyber Storm VIII After-Action Report (2022-08)
https://www.cisa.gov/sites/default/files/publications/FINAL-Cyber-Storm-VIII-After-Action-Report-082022.pdf

S19 - CISA: Cyber Storm IX National Cyber Exercise (2024-04)
https://www.cisa.gov/resources-tools/resources/cyber-storm-ix

S20 - NATO CCDCOE: Locked Shields (current)
https://ccdcoe.org/exercises/locked-shields/

S21 - DOE CESER: Exercises and Training (current)
https://www.energy.gov/ceser/exercises-and-training

S22 - U.S. DOJ: Justice Manual 9-48.000 CFAA (current)
https://www.justice.gov/jm/jm-9-48000-computer-fraud

S23 - DARPA: AIxCC program record ($29.5M cumulative prizes)
https://www.darpa.mil/research/programs/ai-cyber

S24 - NSF: SaTC 2.0 / NSF 25-515 (active page accessed 2026-11-04)
https://www.nsf.gov/funding/opportunities/nsf-satc-us-national-science-foundation-security-privacy-trust/nsf25-515/solicitation

S25 - DHS S&T: Funding and Partnership Opportunities (updated 2026-08-31)
https://www.dhs.gov/science-and-technology/funding-and-partnership-opportunities

S26 - DHS S&T: Small Business Innovation Research Program (current)
https://www.dhs.gov/science-and-technology/sbir

S27 - DARPA: Office-wide Broad Agency Announcements (current)
https://www.darpa.mil/research/opportunities/baa

S28 - SBA: SBIR.gov America's Seed Fund (current)
https://www.sbir.gov/

S29 - U.S. DOJ: 2022 CFAA charging policy announcement (2022-05-19)
https://www.justice.gov/archives/opa/pr/department-justice-announces-new-policy-charging-cases-under-computer-fraud-and-abuse-act

Appendix I - Research Limitations

- This was public-source desk research; classified and non-public operational evidence was unavailable.
- Search results vary in depth. Primary sources were prioritized, but two NIST control summaries are secondary renderings and should be replaced with the official control catalog during final synthesis.
- No live testing, procurement quotation, or stakeholder interview was conducted.
- Cost estimates are planning calculations, not quotes.
- Funding status is time-sensitive and must be rechecked immediately before application.
- No finding in this report constitutes legal advice, deployment authorization, funding commitment, or canon ratification.